

Appendix

GiHyeon Kim, KunHo Heo, SuYeon Kim, and MyeongAh Cho*

Overview. The Appendix is structured as:

- 1) **Appendix §A** Detailed Analysis of Empirical Findings
- 2) **Appendix §B** Implementation Details
- 3) **Appendix §C** Additional Long-Tail Analysis
- 4) **Appendix §D** Limitations and Further Works

A. DETAILED ANALYSIS OF EMPIRICAL FINDINGS

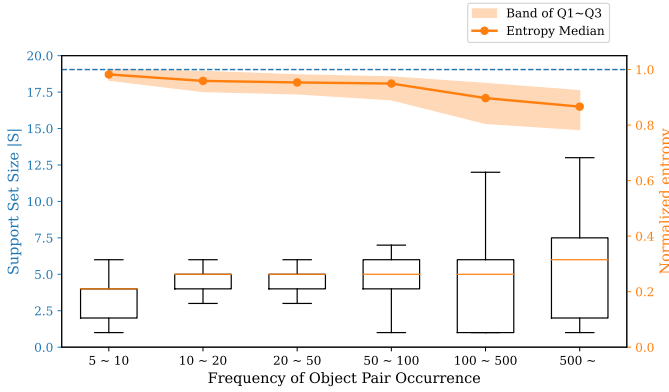


Fig. A1. **Detailed statistical evidence for Sec. III-A.** We report bin-wise statistics over object-pair occurrence frequency. The boxplots (left) show the distribution of the support set size $|S_{ij}|$. The orange line plot (right) shows the median of the normalized predicate entropy $\bar{H}_{ij}$, and the shaded band indicates the inter-quartile range (Q1–Q3), characterizing the variability of predicate distributions within each frequency bin.

In Sec. III-A of the manuscript, we provide statistical evidence supporting the theoretical assumptions introduced in Sec. III-B. Our empirical analysis yields three key observations: (i) improving object representation discriminability alone does not fully resolve the long-tailed predicate issue, (ii) object pairs in scene-graph triplets are themselves long-tailed, and (iii) conditioning on an object pair p_{ij} leads to a substantially flatter predicate distribution $P(r | p_{ij})$ compared to the global predicate prior $P_{\text{data}}(r)$. In this supplementary section, we provide complementary details and clarifications for these findings.

Fig. A1 reports the support-set size $|S_{ij}|$ and the normalized entropy of the conditional predicate distribution for object pairs. To measure the entropy of $P(r | p_{ij})$ meaningfully, we restrict the predicate domain to the *support-set* $S_{ij} = \{r | T_{ijr} > 0\}$, where T_{ijk} denotes the number of triplets in the training set that contain object pair $p_{ij} = \langle o_i, o_j \rangle$ with predicate r . This restriction is necessary because many predicates are structurally infeasible or never observed for a given pair (e.g., predicates like *lower than* which can not be applied to *(table, floor)*). Including such infeasible predicates would artificially inflate the label space and obscure

the actual distribution among feasible relations. We therefore define the conditional distribution over the support as

$$P_{ij}^{\text{supp}}(r) = P(r | p_{ij}, r \in S_{ij}), \quad (1)$$

and compute its entropy as

$$H(P_{ij}^{\text{supp}}) = - \sum_{r \in S_{ij}} P_{ij}^{\text{supp}}(r) \log P_{ij}^{\text{supp}}(r). \quad (2)$$

To compare skewness across pairs with different support sizes, we normalize the entropy by its maximum value $\log |S_{ij}|$:

$$\bar{H}_{ij} = \frac{1}{\log |S_{ij}|}, H(P_{ij}^{\text{supp}}). \quad (3)$$

The results in Fig. A1 support two important conclusions. First, while frequent pairs tend to exhibit larger predicate support sets, the support size $|S_{ij}|$ remains bounded in practice – it rarely exceeds 12 predicates, and for most object pairs it stays around 5. This result indicates that the reported predicate-related statistics are reasonably aligned with our claim, and do not artificially inflate or deflate the entropy with respect to the support set size. Second, even frequency bin with substantial support set variance, the conditional predicate distributions remain consistently high-entropy – the median normalized entropy $\bar{H}_{ij}$ stays close to 1.0 even for frequent pairs. This indicates that once an object pair is given, probability mass is relatively distributed evenly among the feasible predicates rather than concentrating on a few dominant predicates as long-tailed problem. In other words, conditioning on p_{ij} attenuates the dominance of globally frequent predicates and yields a comparatively flattened predicate distribution. Collectively, these findings reinforce our central motivation that predicate imbalance is tightly coupled with **object co-occurrence structure**, and that **object-centric modeling** of pair statistics provides a principled route for mitigating long-tailed predicate problem.

B. IMPLEMENTATION DETAILS

Details on Object Data Collection. To prepare 2D image observations for each 3D instance, we adopt the data selection pipeline of VL-SAT [1], which associates object point clouds with relevant RGB frames via projection. Given an RGB-D stream with camera poses $\{I_1, \dots, I_N\}$, where each frame is represented as $I_t = \langle I_t^{\text{rgb}}, I_t^{\text{depth}}, \mathbf{x}_t \rangle$ (RGB image, depth map, and 6-DoF camera pose), we project the 3D points of an object instance $\hat{o}_i$ into the image plane of I_t^{rgb} using I_t^{depth} and $\mathbf{x}_t$. We then compute the 2D bounding box that encloses the projected points and use its area as a proxy for visibility and coverage. For each object instance, we select **at most** five RGB frames in which projection is successful (i.e., the projected

TABLE A1
SPLITS OF PREDICATES. WE ADOPT THE SAME PREDICATE SPLITS AS VL-SAT [1]. BOTH TAB. IV IN MANUSCRIPT AND FIG. A2 FOLLOW THIS SPLIT SETTING.

Split	Predicate
Head	left, right, front, behind, close by, same as, attached to, standing on
Body	bigger than, smaller than, higher than, lower than, lying on, hanging on
Tail	supported by, inside, same symmetry as, connected to, leaning against, part of, belonging to, build in, standing in, cover, lying in, hanging in

points are valid and yield a non-degenerate bounding box), prioritizing frames with larger projected area. Because objects are not observed consistently across the RGB-D sequence, the number of selected views varies by instance; we denote the resulting number of RGB images for $\hat{o}_i$ as K_i .

Object encoder pre-training. We pre-train the object encoder using PyTorch 1.12 with CUDA 11.3 on a single NVIDIA GeForce RTX 3060 Ti GPU. Optimization is performed with Adam for 100 epochs using a global batch size of 256 and an initial learning rate of 0.01, which is annealed to zero via a cosine schedule. We maintain the official 3DSSG train/validation split throughout pre-training. Model selection is based on validation performance measured by the sum of top- K accuracies for $K \in \{1, 5, 10\}$, where prediction is obtained via CLIP[2]-style maximum similarity matching. Under this configuration, pre-training typically converges within a few hours, and we use the best-performing checkpoint for downstream experiments. To evaluate object classification accuracy independently of the scene-graph model, we further train a linear classifier on frozen object features. This classifier consists of a single fully connected layer trained with AdamW using a batch size of 2048, a fixed learning rate of 1×10^{-4} , and standard cross-entropy loss, without additional learning-rate scheduling.

Scene graph prediction. All 3D semantic scene graph experiments are conducted in PyTorch 1.12 (CUDA 11.3) on a single NVIDIA GeForce RTX 3090 GPU. We train the model for 150 epochs using AdamW with a global batch size of eight scenes. The learning rate is initialized to 1×10^{-4} and decayed using a cosine schedule. For each object instance, we randomly down-sample the associated point cloud to 256 points. Training typically converges within approximately 72 hours. We select the checkpoint that achieves the highest validation mean Recall (mR) for SGCLs at R@20, and report all quantitative and qualitative results using this checkpoint. All other hyperparameters are followed by original settings on our previous work [3].

C. ADDITIONAL LONG-TAIL ANALYSIS

As shown in Fig. A2, we visualized the predicate frequency in the training set as a line chart and report the class-wise mR@1 as a bar chart for each predicate. The frequency curve clearly exhibits a long-tailed distribution, where only a few head predicates occur very frequently while most predicates appear rarely. The specific predicates belonging to the Head/Body/Tail splits are listed in Tab. A1. In this setting,

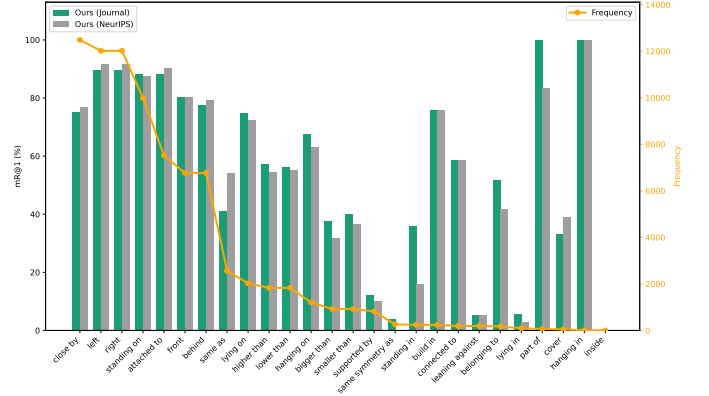


Fig. A2. **Classwise Top-1 recall of predicate.** We reported mR@1 of each predicate class compared to our previous baseline [3]. The line shows predicate frequency in the training dataset of 3DSSG [4] and predicates(x-axis) are ordered by frequency .

Ours (Journal) consistently improves class-wise mR@1 on the body and tail predicates, indicating stronger robustness to data imbalance. Although **Ours (Journal)** shows a minor drop on some head predicates, it achieves notable gains on tail predicates, which are typically the most challenging and most critical for improving mR@1. In particular, **Ours (Journal)** yields substantial improvements on tail predicates such as part of, belonging to, and standing in, demonstrating its effectiveness in enhancing long-tail relational recognition. However, the head predicate same as exhibits a noticeable performance drop, suggesting that our method still has limitations in handling certain frequent predicates.

D. LIMITATIONS AND FURTHER WORKS

Our study strictly follows the evaluation protocol of 3DSSG and therefore does not consider the open-vocabulary setting that has recently gained momentum with the emergence of LLMs and LVLMs. We intentionally focus on the closed-vocabulary regime to conduct controlled cause-and-effect analyses, isolating the impact of object-centric representations on long-tailed predicate reasoning. Consequently, our method is not directly comparable to, nor immediately compatible with, open-vocabulary 3D scene graph generation pipelines that rely on LLM/LVLM-based prompting or decoding. This limitation restricts our ability to handle previously unseen object categories or relationship predicates at inference time.

As future work, we aim to relax the closed-vocabulary assumption and investigate how LLMs/LVLMs can be integrated with our object-centric framework to infer more precise and compositional relationships in an open-vocabulary setting. Notably, our results suggest that object-pair signals are a strong inductive cue for scarce-predicate reasoning and relationship generalization, which could serve as a reliable interface between geometric perception and language-based reasoning. We believe this object-centric perspective provides a principled foundation for scaling 3D scene graph prediction beyond fixed label sets, enabling more accurate and efficient reasoning when combined with LLM/LVLM capabilities. Accordingly, our subsequent research will explore effective strategies for

open-vocabulary object-centric 3D scene graph prediction, including language-grounded relation discovery and robust generalization to novel classes.

REFERENCES

- [1] Z. Wang, B. Cheng, L. Zhao, D. Xu, Y. Tang, and L. Sheng, “VI-sat: Visual-linguistic semantics assisted training for 3d semantic scene graph prediction in point cloud,” in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2023, pp. 21 560–21 569.
- [2] A. Radford, J. W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark *et al.*, “Learning transferable visual models from natural language supervision,” in *International conference on machine learning*. PmLR, 2021, pp. 8748–8763.
- [3] K. Heo, G. Kim, S. Kim, and M. Cho, “Object-centric representation learning for enhanced 3d semantic scene graph prediction,” in *The Thirty-ninth Annual Conference on Neural Information Processing Systems*, 2025.
- [4] J. Wald, H. Dhano, N. Navab, and F. Tombari, “Learning 3d semantic scene graphs from 3d indoor reconstructions,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2020, pp. 3961–3970.